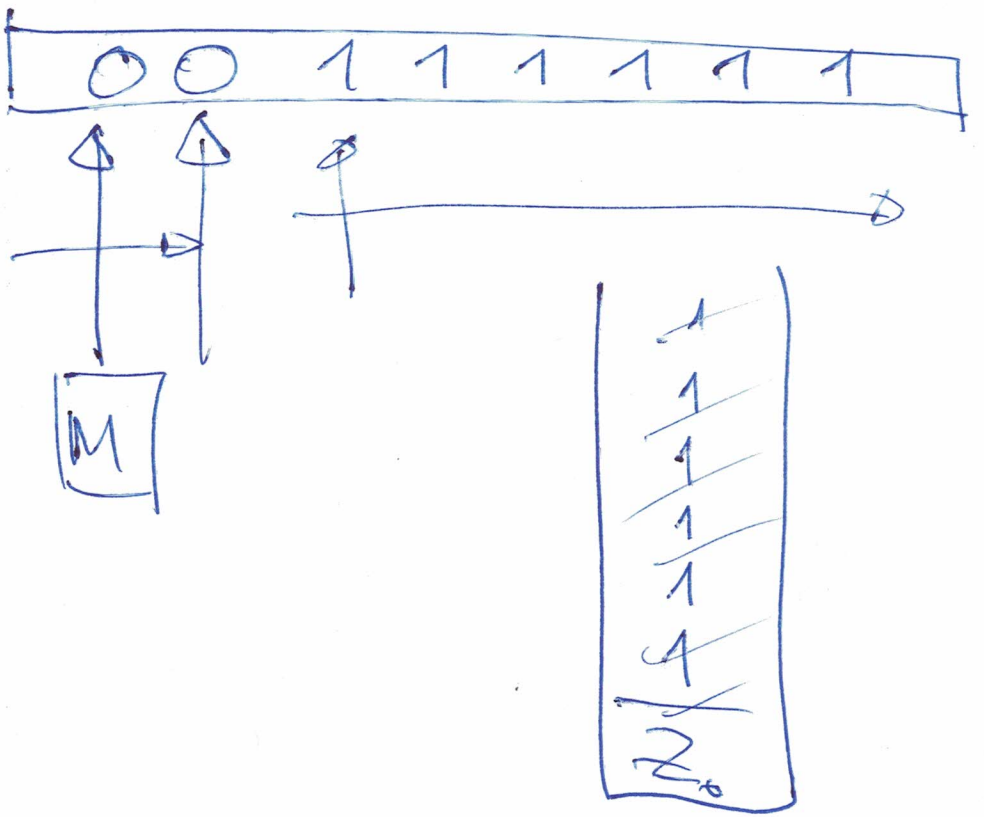
Cpts 317. 11/7

hw hint?

Prob 1. $L = \{0^n 1^{3n} : n \geq 1\}$.



$\delta(q_0, 0, Z_0) = (q_1, 111 Z_0)$

$\delta(q_1, 0, 1) = (q_1, \ldots\ldots)$

$\vdots$

$\delta(q, \ldots, Z_0$

Prob 2. $\mathcal{L} = \{ w \in \{0,1\}^* : \#_0(w) \geq \#_1(w) \}$.

look at your class notes.

key: I need two modes:

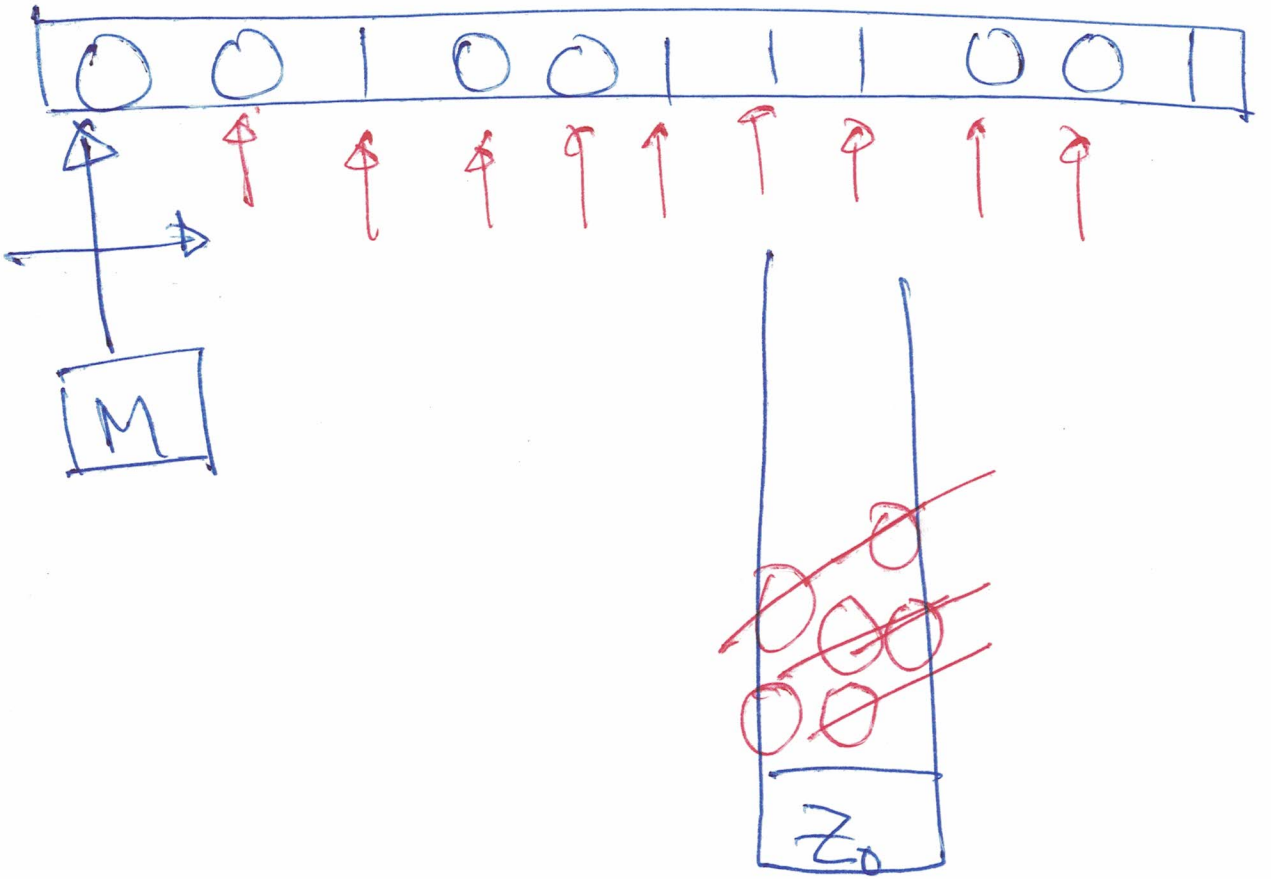
mode 1: read 0 $\longrightarrow$ push 0

read 1 $\longrightarrow$ pop 0

mode 2: read 1 $\longrightarrow$ push 1

read 0 $\longrightarrow$ pop 1.

Prob 3. $\mathcal{L} = \{ w \in \{0,1\}^* :$ each prefix of w sat $\#_0 \geq \#_1 \}$.



Prob 1. $\delta(q_0, 0, Z_0) = (q_1, 111 Z_0)$;

$\delta(q_1, 0, 1) = (q_1, 1111)$;

$\delta(q_1, 1, 1) = (q_2, \Lambda)$

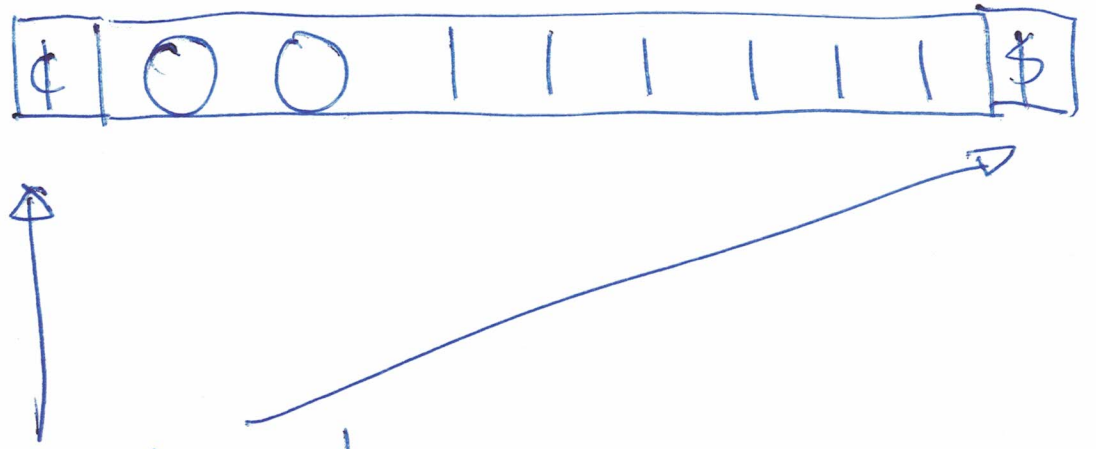
$\delta(q_2, 1, 1) = (q_2, \Lambda)$

$\delta(q_2, \Lambda, Z_0) = (q_3, Z_0)$

accepting.

// assuming that input has end markers.
//

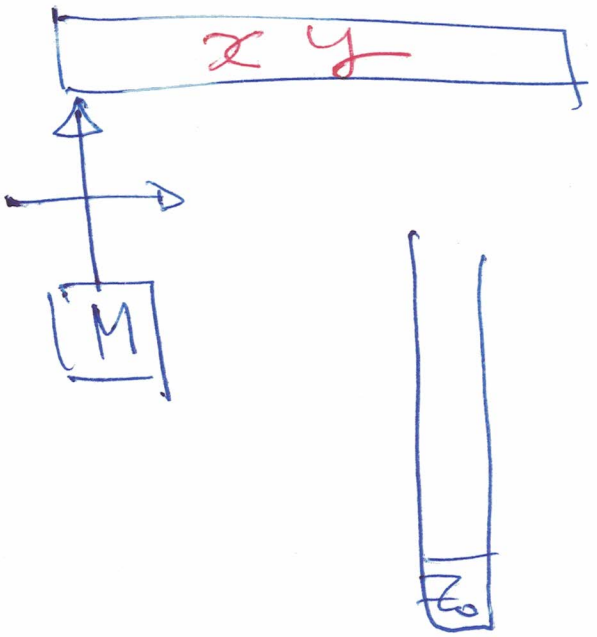
good old days:



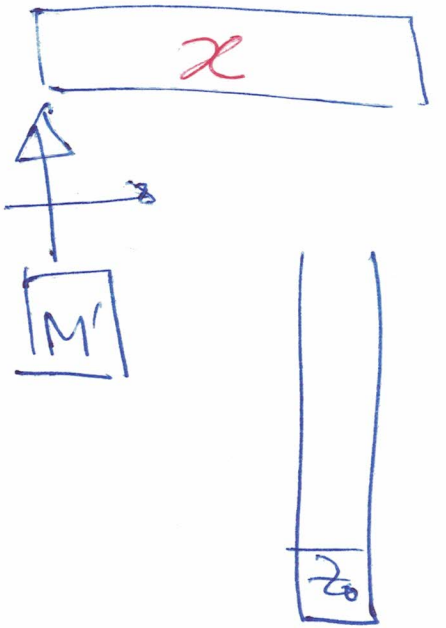
end markers.

Prob 4.

Given: L is accepted by M.



Want: M' accepts $\text{Prefix}(L)$
$= \{x : \exists y.\ xy \in \mathcal{L}\}$.



M accepts xy, iff M' accepts x

you don't know what the y is.

Thm. (Pumping Lemma). Let $\mathcal{L}$ be an inf. c.f. language. Then there is a positive integer n such that for each word $z \in \mathcal{L}$ with $|z| \geq n$, z can be written into 5 parts:

$$z = uvwxy$$

for some u, v, w, x, y sat:

① $|vwx| \leq n$

② $|vx| \geq 1$

③ $\forall i \geq 0$, $uv^iwx^iy \in \mathcal{L}$.

Proof. Hard.

Show $\mathcal{L} = \{a^n b^n c^n : n \geq 0\}$ is not c.f.

Proof. (use P.L). Assume that $\mathcal{L}$ is c.f. According to P.L., let x be the number in the Lemma.

Pick $z = a^n b^n c^n$ with $|z| = 3n > x$.

for each possible way to write z into 5 parts:

$z = uvwxy$

Sat: $|vwx| \leq n$, $|vx| \geq 1$, we can see:

the picked z must be in $\mathcal{L}$, must dep. on x.

$$z = \overbrace{a \cdots a}^{\leftarrow n \rightarrow}\,\overbrace{b \cdots b}^{\leftarrow n \rightarrow}\,\overbrace{c \cdots c}^{\leftarrow n \rightarrow}$$

$$z = \underbrace{\boxed{u}\,\boxed{v}\,\boxed{w}\,\boxed{x}\,\boxed{y}}_{\leftarrow\ \leq n\ \rightarrow}$$

The middle three blocks combined vwx can contain at most two kinds of symbols; therefore, since $|vx| \geq 1$, we have, taking $i=2$,

$uvvwxxy \notin L$ since the three symbol counts are off-balanced. $\blacksquare$

Closure properties of c.f. languges.

Recall: reg languges are closed under all Boolean operators: $\cap$, $\cup$, $\overline{}$, Kleene star.

complement.

Thm. c.f. languges are closed under $\cup$.

// if $\mathcal{L}_1$ and $\mathcal{L}_2$ are c.f., then so is $\mathcal{L}_1 \cup \mathcal{L}_2$.

//

Proof. (by Exmple).

Given two c.f. grammars:

first grammar:

$$S \to ASB \mid ab$$
$$A \to Aa \mid \Lambda$$
$$B \to Bb \mid \Lambda$$

G_1

Second grammar:

$$S \to aA$$
$$A \to aAb \mid a.$$

G_2

Want: a c.f. grammar to accept $\mathcal{L}(G_1) \cup \mathcal{L}(G_2)$

new grammar:

$$S \to S_1 \mid S_2$$
$$S_1 \to A_1 S_1 B_1 \mid ab$$
$$A_1 \to A_1 a \mid \Lambda$$
$$B_1 \to B_1 b \mid \Lambda$$

$$S_2 \to aA_2$$
$$A_2 \to aA_2b \mid a.$$

Thm. c.f. languages are NOT closed under $\cap$.

Proof. From P.L., we know that

$L = \{a^n b^n c^n : n \geq 0\}$ is not c.f. Then, the result follows since

$$L = \{a^n b^m c^m : n, m \geq 0\} \cap \{a^m b^m c^n : n, m \geq 0\}$$

is the intersection of two c.f. languages.

Thm. c.f. languages are NOT closed under complement.

Proof. $L_1 \cap L_2 = \overline{(\overline{L_1} \cup \overline{L_2})}$, the result follows.

Decision queries on c.f. languages:

<u>membership</u>: Given: a PDA M, a word w

Question: $w \in L(M)$?

<u>emptiness</u>: Given: a PDA M

Question: $L(M) = \emptyset$?

We have algs to solve membership &
solve emptiness.